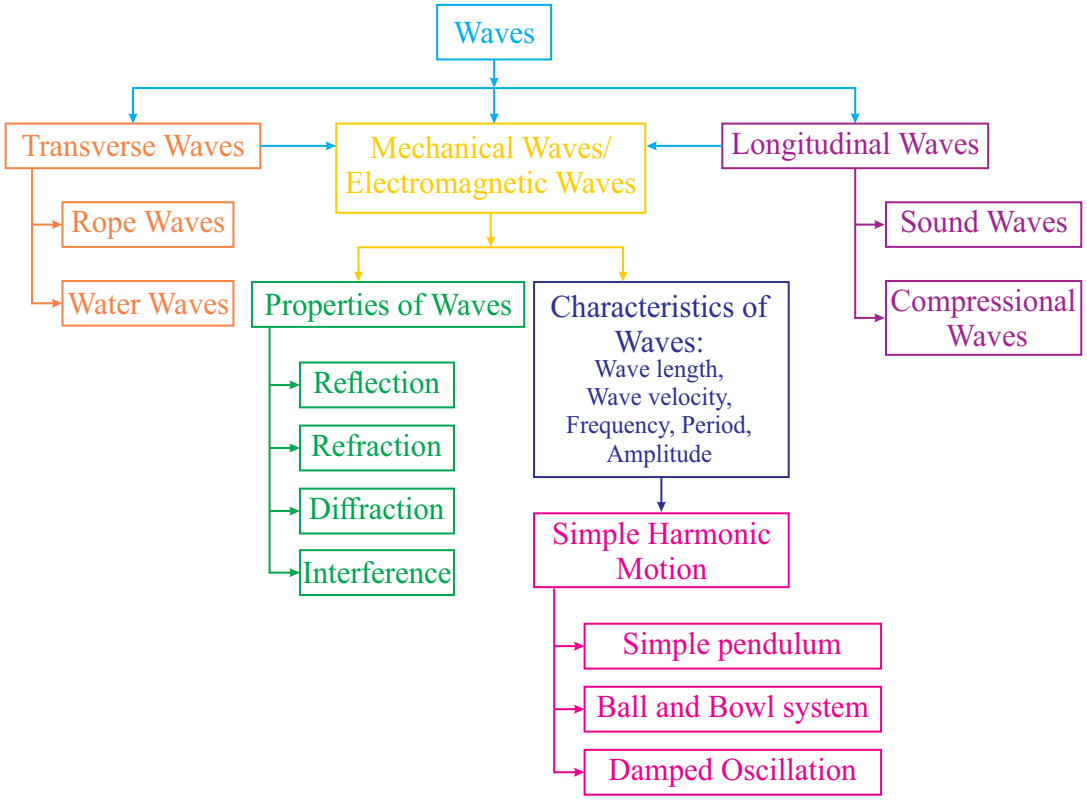
CONCEPT MAP



Section (A) Multiple Choice Questions (MCQs)

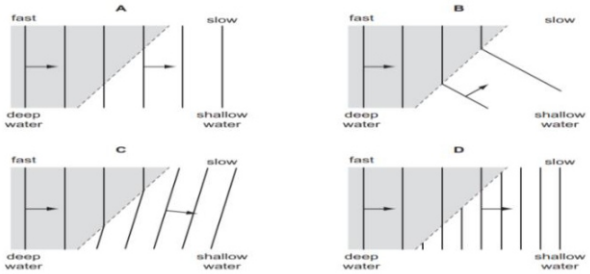
1. A girl throws a small stone into a lake. Waves spread out from where the stone hits the water and travel to the bank of the lake. She notices that ten waves reach the side of the pond in a time of 5.0s. What is the frequency of the waves?

a) 0.50Hz b) 15Hz
c) 2.0Hz d) 50Hz

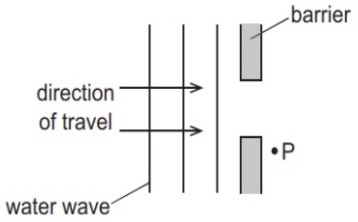
2. Water waves can be used to show reflection, refraction, and diffraction. For each of these, which row shows whether or not the speed of the water waves changes?

	Reflection	Refraction	Diffraction
a)	Yes	yes	Yes
b)	Yes	No	yes
c)	No	yes	no
d)	No	no	no

3. The diagrams show water waves that move more slowly after passing into shallow water. Which diagram shows what happens to the waves?



4. The diagram shows a water wave approaching a barrier with a gap.



The wave reaches point P. What is the name of the effect that causes the wave to reach point P?

a) Diffraction b) Dispersion
c) reflection d) Refraction

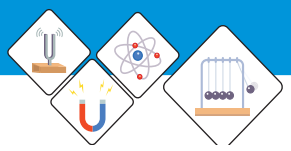
5. Water waves pass from deep into the shallow region then refracted. The characteristics of wave which will remains constant is:

a) Direction b) Frequency
c) Speed d) Wavelength.

6. Which is not a characteristic of wave?

a) An amplitude b) Period
c) Mass d) Velocity

7. When an oscillating object is in simple harmonic motion, its maximum speed occurs when the object is at its



a) Highest point. b) Lowest point
c) Equilibrium point d) Extreme point

8. In an oscillating pendulum, the bob accelerates from its extreme position due to
a) Inertia b) Tension in the string
c) Wind d) Gravitational force

9. In the ball and bowl system, the mean position is at
a) The earth b) Floor of the bowl
c) Center of bowl d) Extreme position

10. Oscillations are damped due to the presence of
a) Linear motion b) Restoring force
c) Frictional force d) Mechanical force

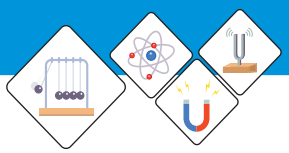
Section (B) Structured Questions

1. Define the term transverse wave.
2. Define the term longitudinal wave.
3. **a)** Write a short note on the mechanical wave.
b) How can you say that mechanical waves are also material waves?
4. Waves are the means of energy transfer without matter. Justify this statement with the help of everyday life examples.
5.. **a)** Define the following terms of a wave:
i) amplitude ii) period
iii) frequency iv) wavelength
b) Derive the formula of wave speed, $v = f\lambda$
6. **a)** What is a ripple tank, and explain its working?
b) Define the wavefront?
7. Reference an experiment to explain the refraction of waves concerning the ripple tank.
8. What is the phenomenon of diffraction?
9. **a)** What is simple harmonic motion?
b) What are the necessary conditions for a body to execute simple harmonic motion?
10. **a)** With the help of a diagram, explain SHM in the pendulum.
b) The period of simple pendulum executing the formula gives SHM

$$T = 2\pi\sqrt{\frac{L}{g}}$$

What will be the effect of the period if there is an increase in its
i) length ii) mass.

11. **a)** With the help of a diagram, explain SHM in the ball and bowl system.



Section (A) Multiple Choice Questions (MCQs)

Choose the correct answer from the following choices:

1. Sound is a form of:
   a) Electrical energy b) Mechanical energy
   c) Thermal energy d) Chemical energy

2. Audible frequencies range that a normal human ear can detect is:
   a) 10 Hz to 10 kHz b) 20 Hz to 20 kHz
   c) 25 Hz to 25 kHz d) 30 Hz to 30 kHz

3. The approximate value of the speed of sound in air at 0°C temperature is:
   a) 331 m / s b) 34 m / s
   c) 17 m / s d) 680 m / s

4. Sound travel faster in solid as compare to gases because of:
   a) Gas molecules are packed loosely.
   b) Sound does not travel faster through a solid than a gas.
   c) Solid molecules are packed tightly.
   d) Gas molecules move faster.

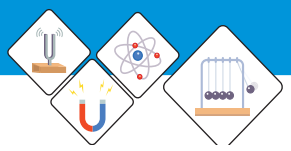
5. The two factors that affect the speed of sound in air are:
   a) Humidity and volume of the air
   b) Temperature and mass of the air
   c) Volume and mass of the air
   d) Temperature and humidity of the air

6. The separation between two consecutive compressions of the sound wave is called:
   a) Time period b) Amplitude
   c) Frequency d) Wavelength

7. The order of speed of the sound in different mediums from faster to slowest is
   a) Gas → Liquid → Solid b) Liquid → Solid → Gas
   c) Solid → Liquid→ Gas d) Gas → Solid → Liquid

8. Ultrasound has several uses in medicine and industry. Which one has use of ultrasound?
   a) Absorption b) Pre-natal scanning
   c) Dispersion d) Measuring humidity of air

9. The causes of the echo is:
   a) Absorption b) Dispersion
   c) Reflection d) Refraction



10. Which type of wave cannot travel through a vacuum?
    a) Sound waves
    b) Infra-red radiation
    c) Microwaves
    d) X-rays

Section (B) Structured Questions

1. a) How is the sound produced?
b) With the help of a diagram, describe how compressions and rarefactions are produced in the air near a source of the sound.

2. a) Why are sound waves referred to as mechanical waves?
b) Sound requires a material medium for its propagation. Cite an experiment to prove this statement.

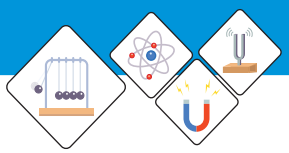
3. a) Distinguish between musical sound and noise.
b) Explain how noise is harmful to humans?

4. a) Define the quality or timbre of the sound.
b) Is it possible that two or more waves from different musical instruments combine to form a single wave?

5. a) Why is the speed of the sound greater in solids than in liquids or gases?
b) Explain the effect of the following factors on the speed of sound in the air.
   i. Temperature
   ii. Humidity

6. a) Define echo.
b) Explain the working and application of a sonar.
c) How can defects in a metal block be detected using ultrasound? Explain with the help of a diagram.

7. a) Define the following terms
   i. Infrasonic
   ii. The audible frequency range of hearing
   iii. Ultrasound
b) How is ultrasound used for cleaning?
c) Explain two applications of ultrasound that are used in hospitals for medical imaging.



Section (A) Multiple Choice Questions (MCQs)

Choose the correct answer from the following choices:

1. The waves that have maximum penetrating power to treat tumors are;
a) Ultraviolet radiation b) Microwaves
c) Gamma-rays d) Radio waves

2. The electromagnetic rays used in radiotherapy to destroy cancer cells are;
a) Infrared rays b) Visible rays
c) X - rays d) Ultraviolet rays

3. The Velocity of light in a diamond is
(whereas the refractive index of a diamond with respect to vacuum is 2.5)
a) 1.2×10^{8} m/s b) 5×10^{8} m/s
c) 1.2×10^{10} m/s d) 2.5×10^{8} m/s

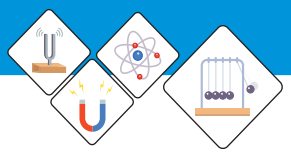
4. The group containing only electromagnetic waves is;
a) Light waves, Radio waves, Microwaves
b) Light waves, Radio waves, Sound waves
c) Light waves, Sound waves, Microwaves
d) Radio waves, Sound waves, Microwaves

5. The list that shows electromagnetic waves in order of an increasing wavelength is;
a) Microwaves, X-rays, Gamma-rays
b) Microwaves, Gamma-rays, X-rays
c) X-rays, gamma-rays, Microwaves
d) Gamma-rays, X-rays, Microwaves

6. The type of electromagnetic wave used in security scanners at night is;
a) Infra-red b) Microwaves
c) Radio waves d) X-ray

7. A narrow beam of white light passes from air into the glass and is refracted. The wave characteristic remains unchanged in its;
a) Direction b) Frequency
c) Speed d) Wavelength

8. The type of waves that are used in the television remote controllers;
a) Radio waves b) Infra-red waves
c) Ultra-violet waves d) Visible light



9. The color that is least deviated by a prism;
   a) Violet ray b) Green ray
   c) Red ray d) Yellow ray

10. The optical phenomenon in which the splitting of white light into seven distinct colors occur is called;
   a) Refraction b) Reflection
   c) Dispersion d) Diffraction

Section (B) Structured Questions

1. **a)** Define dispersion of light.
   b) Describe the dispersion of light when passing through a glass prism.

2. **a)** Explain how the rainbow is produced on a rainy day?
   b) Explain how the colors are related to distinct frequency or wavelength?

3. **a)** What are electromagnetic waves?
   b) List the main components of the electromagnetic spectrum in decreasing order of their wavelengths.
   c) Ultraviolet rays have a higher frequency than radio waves. Can UV rays travel faster in a vacuum?

4. Compare the properties of ultraviolet rays and radio signals.
   a) Which one travels at a faster speed?
   b) Which wave has a greater frequency?
   c) Which wave has a greater wavelength?

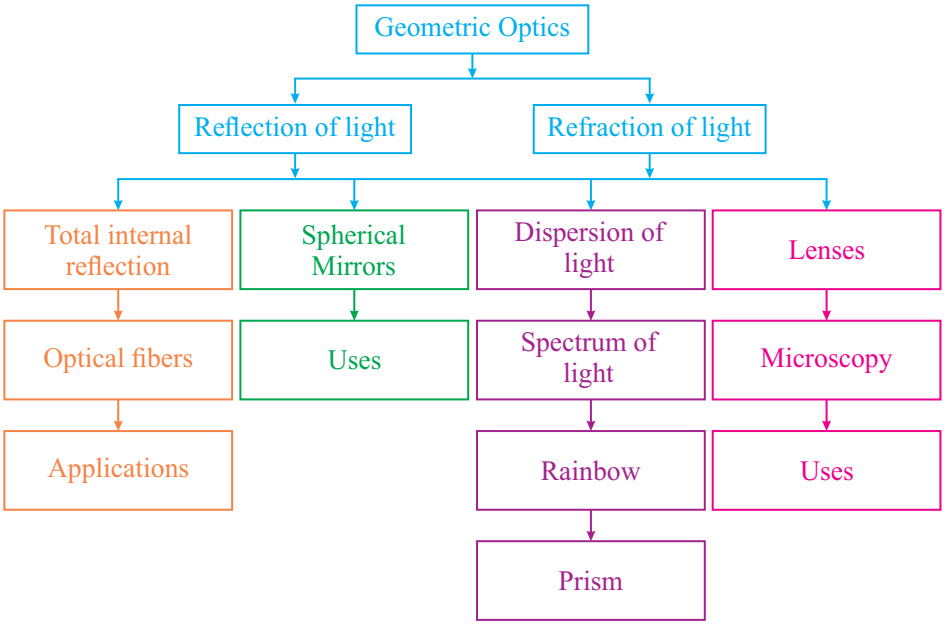
5. **a)** What are the main sources of radio waves?
   b) What is the main advantage of using radio waves in communication?

6. Why are microwaves preferred in satellite communication?

7. **a)** What type of radiation is commonly used in remote controllers for household appliances?
   b) How do the molecules emit infrared radiations?
   c) How intruder alarms help security personnel visualize the thermal images

8. **a)** On what principle do optical fibers work?
   b) Reference the daily life applications of optical fibers in;
   i. telecommunication
   ii. medical industry?

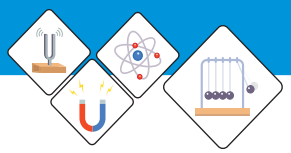
CONCEPT MAP



Section (A) Multiple Choice Questions (MCQs)

Choose the correct answer from the following choices:

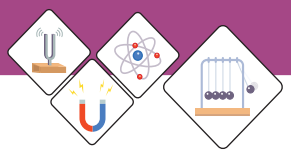
1. In a concave mirror, the image size depends upon
   (a) Size of the object (b) Position of the object
   (c) Area covered by the object (d) The shape of the object
2. In the normal human eye, the image is formed
   (a) In from of the retina (b) Behind the retina
   (c) On the retina (d) In between lens and retina
3. When a light ray enters from a denser medium to a rare medium, it bends
   (a) Perpendicular to normal (b) Parallel to normal
   (c) Toward normal (d) Away from normal
4. In a compound microscope, as compared to an objective, the eyepiece lens has a focal length
   (a) Zero (b) negative
   (c) Small (d) Large
5. When the angle of refraction is 90° and the refractive index for water is 1.33, the critical angle is
   (a) 48.8° (b) 49.1°
   (c) 50.0° (d) 51.0°



6. To view dim stars, we use
   (a) Compound microscope (b) Simple microscope
   (c) Endoscope (d) Telescope
7. The human eye acts like a
   (a) Camera (b) Projector
   (c) Telescope (d) Microscope
8. A magnifying glass forms an enlarged
   (a) Real and upright image (b) Real and inverted
   (c) Virtual and upright image (d) Virtual and inverted image
9. The entire light is reflected into the same denser medium, which is called total
   (a) External reflection (b) Internal reflection
   (c) External refraction (d) Internal refraction
10. In the optic fiber, the core is made of glass or plastic of relatively
   (a) Zero refractive index (b) High refractive index
   (c) Low refractive index (d) No refractive index
11. A magnifying glass is also called
   (a) Endoscope (b) Compound microscope
   (c) Simple microscope (d) Telescope
12. The defect in which the image is formed beyond the retina is called,
   (a) Long-sightedness (b) Short-sightedness
   (c) Blind spotting (d) Image defect
13. The short-sightedness can be corrected by using.
   (a) Convex glasses (b) Convex mirror
   (c) Concave mirror (d) Convex glasses
14. Lenses form images through
   (a) Dispersion (b) Refraction
   (c) Diffraction (d) Reflection
15. To illuminate the inaccessible places in the tooth, dentists use
   (a) Concave mirror (b) Convex mirror
   (c) Convex lens (d) Concave lens

Section (B) Structured Questions

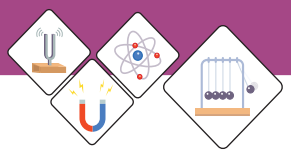
1. **a.** What do you understand by the term reflection of light?
   b. Outline a diagram to illustrate reflection at a plane surface.
   c. Describe the following terms used in reflection:
   i. Normal **ii.** The angle of incidence **iii.** The angle of reflection.
   d. Also, express the laws of reflections.
2. Name the type of mirror used in the following situations.
   a. Side /rearview mirror of a vehicle.



Section (A) Multiple Choice Questions (MCQs)

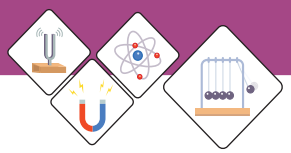
Choose the correct answer from the following choices:

1. Branch of physics which deals with the charges at rest is called
   (a) Electricity (b) Electrostatic
   (c) Quantum (d) Magnetism
2. The magnitude of force between two unit positive charges when the distance between them is 1m would be
   (a) 0 N (b) 1N (2) 2N (d) Coulombs Constant
3. Coulombs law most closely resembles with
   (a) Law of conservation of Energy (b) Newton's Law of Gravitation
   (c) Newton's 2nd law of Motion (d) Faraday's law
4. If the electrostatic force between two electrons is ***F*** Newton, then the electrostatic force between two protons at the same distance is
   (a) 0N (b) 2F (c) 2/3 F (d) F
5. The direction of electric force and electric field intensity is
   (a) Parallel to each other (b) Perpendicular to each other
   (c) opposite to each other (d) In any direction
6. The work done on a unit charge against electric field intensity is called
   (a) Electric field (b) Electric current
   (c) Electric potential (d) electric field
7. The capacitance of capacitors increases when they connected in
   (a) Parallel (b) Series (c) both (d) none of them
8. Two capacitors of 8μF are connected in series then the equivalent capacitance is
   (a) ¼ μF (b) 2 μF (c) 4 μF (d) 6 μF
9. The presence of a dielectric between the plates of capacitors, the capacitance of capacitor
   (a) Increases (b) decreases
   (c) remain constant (d) remain uncharged
10. If the area of the parallel plate capacitor is doubled then the capacitance will be
    (a) Remain uncharged (b) half
    (c) Double (c) increased two times



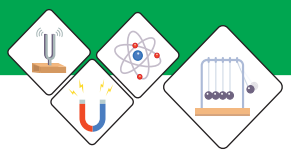
Section (A) Multiple Choice Questions (MCQs)

1. In an Electric circuit when Electrons move from low to high potential they will:
   (a) Gain Energy (b) Lose their identity
   (c) Lose Energy (d) Gain Potential
2. In an electric circuit an ammeter is always connected in
   (a) Series (b) parallel
   (c) mixed (d) none of the above
3. Resistance of a conductor does not depend on
   (a) Length of the conductor (b) Area of crossection
   (c) Density (d) Resistivity
4. Ohm's law states that:
   (a) Resistance increases as current increases
   (b) Resistance decreases as current increases
   (c) Resistance increases as voltage increases
   (d) Current increases as voltage increases
5. The condition when the resistance of a circuit is zero is known as
   (a) Closed-circuit (b) Open circuit
   (c) Short circuit (d) Zero circuit
6. The condition for the validity of Ohm's law is that the
   (a) Temperature should remain constant
   (b) Current should be proportional to voltage
   (c) Resistance must be wire wound type
   (d) All of the above
7. Ohm's law is not applicable to
   (a) Semiconductors (b) D.C. circuits (c) Small resistors (d) High currents
8. Two resistances of 6 Ω and 12 Ω are connected in parallel. Their net resistance is ____.
   (a) 7Ω (b) 6 Ω (c) 4 Ω (d) 5 Ω
9. The property of a body to oppose the flow of electric charge through it is called electric _____.
   (a) Capacitance (b) potential
   (c) resistance (d) conductance
10. Which of the following is the purpose of connecting a battery in an electric circuit?
    (a) To maintain resistance across the conductor.
    (b) To vary resistance across the conductors.
    (c) To maintain constant potential difference across the conductor.
    (d) To maintain varying potential difference across the conductor.



Section (A) Multiple Choice Questions (MCQs)

1. Which statement is true about the magnetic poles?
   (a) unlike poles repel
   (b) like poles attract
   (c) magnetic poles do not effect each other
   (d) a single magnetic pole does not exist
2. What is the direction of the magnetic field lines inside a bar magnet?
   (a) from north pole to south pole (b) from south pole to north pole
   (c) from side to side (d) there are no magnetic field lines
3. The presence of a magnetic field can be detected by a
   (a) small mass (b) stationary positive charge
   (c) stationary negative charge (d) magnetic compass
4. If the current in a wire which is placed perpendicular to a magnetic field increases, the force on the wire
   (a) Increases (b) decreases
   (c) remains the same (d) will be zero
5. A D.C motor converts
   (a) mechanical energy into electrical energy
   (b) mechanical energy into chemical energy
   (c) electrical energy into mechanical energy
   (d) electrical energy into chemical energy
6. Which part of a D.C motor reverses the direction of current through the coil every half-cycle?
   (a) the armature (b) the commutator
   (c) the brushes (d) the slip rings
7. The direction of induced e.m.f. in a circuit is in accordance with conservation of
   (a) Mass (b) charge (c) momentum (d) energy
8. The step-up transformer
   (a) increases the input current
   (b) increases the input voltage
   (c) has more turns in the primary
   (d) has less turns in the secondary coil
9. The turn ratios of a transformer is10. It means
   (a) $I_S = 10\ I_p$ (b) $N_p = 10 N_S$
   (c) $N_S = 10\ N_p$ (d) $V_S = 10 V_p$



Section (A) Multiple Choice Questions (MCQs)

1. Metals are good conductors of electricity because they have free:
   (a) Electrons (b) Protons (c) Neutrons (d) Photons
2. The continuous flow of electrons is made possible by a device called:
   (a) Cathode (b) Electron gun (c) Anode (d) Filament
3. The electric field can be detect:
   (a) Photon (b) Neutron (c) Proton (d) Electron
4. If the direction of magnetic field is reversed, the direction of force is:
   (a) Reversed (b) Not reversed (c) May or may not reversed (d) None of these
5. The process of emission of electrons from the hot metal surfaces is called
   (a) Plastic emission (b) Thermionic emission
   (c) Static emission (d) Current emission
6. If input of a NOT gate is "1" then its output is:
   (a) 1 (b) 0 (c) may be 1 or may be 0 (d) None of these
7. The Boolean expression of an AND gate is:
   (a) A . B (b) A + B (c) A × B (d) None of these.
8. Electronics comprises the:
   (a) Physics (b) Engineering (c) Technology (d) All of these
9. The Boolean expression of an OR gate is:
   (a) A . B (b) A + B (c) A − B (d) None of these.
10. The cathode ray carry
    (a) positive charge (b) neutral (c) negative charge (d) positrons

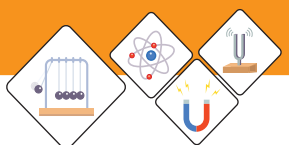
Section (B) Structured Questions

1. Give an example showing that the world is shifting from low-tech electrical appliances to high-tech electronic appliances
2. (a) Write any three advantages of Digital electronics (devices) over analogue electronics devices.
   (b) Define the role of vacuum tube in electronics.
3. What is the function of a 'DAC'?
4. What makes the cathode give off electrons?
5. (a) Demonstrate the process of thermionic emission by diagram?
   (b) Will the process of thermionic emission still occurs, if air is in the tube instead of having vacuum in it?
6. Why the cathode repel electrons?
7. Write any two properties of cathode rays.



Section (A) Multiple Choice Questions (MCQs)

1. Another name for a supercomputer is a:
   a) High-performance computer b) Maxi computer
   c) Mainframe computer d) None
2. Input, processing, output. and storage are collectively referred to as:
   a) Information processing cycle. b) Software life cycle.
   c) Hardware life cycle. d) Information technology.
3. _____is the output from a computer that ranks from processing input data
   a) Data b) Information
   c) Computer d) Mouse
4. Which one of the following is not considered as a system software?
   a) Assembler b) Interpreter
   c) Compiler d) Tally
5. Which of the following is suitable for connecting different computers in an organized manner within an office building?
   a) MAN b) WAN
   c) ANN d) LAN
6. A computer program that translates one program instruction at a time into machine language is called?
   a) Interpreter b) CPU
   c) Compiler d) Simulator
7. The name given to a sequence of instructions in a computer language, to get the desired result is?
   a) Program b) Decision table
   c) Pseudo code d) Algorithm
8. USB stands for
   a) Ultra Serial Bus b) Unlimited Structured Bit
   c) Universal Serial Bus d) Unified Status Bus
9. Which is the extension not suitable to an ms-word file
   a) .doc b) .docx
   c) .rtf d) .jpeg
10. ICT stands for
    a) Information and Communications Technology
    b) Integrated Circular Technology
    c) Intensive Computer Techniques
    d) Interfacing Computer Theories



SUMMARY

- Everything around us is made up of atoms.
- Atom is the smallest unit of matter.
- Every atom is composed of two parts; the nucleus and the shell.
- The nucleus is a central small and dense part of the atom that contains protons and neutrons.
- The shell part of the atom contains electrons that orbit at high speed around the nucleus.
- Most of the atom is an empty space.
- In an atom, electric forces bind the electrons to the nucleus.
- An atom consists of three elemental particles: electrons, protons, and neutrons.
- Each element, however, does have a unique number of protons in its nucleus.
- The number of protons in an atom of an element is called the atomic number.
- The number of protons and neutrons in an atom is known as nucleons.
- The total number of nucleons in an atom is called the mass number.
- Isotope has the same atomic number but a different mass number.

Section (A) Multiple Choice Questions (MCQs)

Choose the correct answer from the following choices:

1. $^{2}_{1}H$ and $^{3}_{1}H$ are:

a) Isotopes b) Isobars
c) Isotones d) Isochores

2. The neutral atoms of all of the isotopes of the same element have

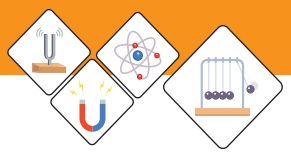
a) different numbers of protons.
b) exact numbers of neutrons.
c) An exact number of protons.
d) An exact number of nucleons.

3. Consider the species $_{17}Cl^{35}$, and $_{17}Cl^{37}$. These species have:

a) the exact number of nucleons
b) the exact number of protons
c) the exact number of neutrons.
d) the exact mass number.

4. Atomic mass of an element is equal to

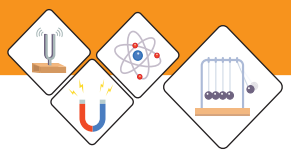
a) Mass of protons and neutrons
b) Mass of protons and electrons
c) Mass of electrons and neutrons
d) Mass of protons only



5. The maximum mass of an atom is concentrated in:
   a) nucleus b) neutrons
   c) protons d) electrons
6. Consider isotope $_{92}U^{237}$ of uranium. The number of neutrons in it is:
   a) 92 b) 237
   c) 145 d) 329
7. The symbol denotes the proton number is:
   a) P b) A
   c) N d) Z
8. The number of neutron(s) in Protium is:
   a) no b) one
   c) two d) three
9. In an atom, the nucleus when compared to the extra-nuclear part, is
   a) More significant in volume and heavier in mass
   b) smaller in volume but heavier in mass
   c) More significant in volume but lighter in mass
   d) Smaller in volume and lighter in mass
10. If an element B has five protons and six neutrons what will be the symbol of element B
   a) $_{11}B^{6}$ b) $_{11}B$
   c) $_{6}B^{11}$ d) $^{5}_{11}B$

Section (B) Structured Questions

1. **(a)** Which particles are found in the nucleus of an atom?
   (b) Describe the structure of an atom.
   (c) How does the number of protons in a nucleus distinguish one element from the other?
2. **(a)** Cite the Geiger Marsden experiment with the help of a diagram.
   (b) Give the Rutherford model of an atom.
   (c) Why it was proposed that most atoms possess an empty space.
3. **(a)** Define Atomic number (Z)
   (b) Explain the symbolic representation of an atom of an element. Give an example.
4. **(a)** What is the isotope?
   (b) Explain the isotope with an example.
   (c) Why are the chemical properties of an element's different isotopes identical?
   (d) List the physical properties of different isotopes of an element that are different.



Section (A) Multiple Choice Questions (MCQs)

Choose the correct answer from the following choices:

1. The α-radiation is
   (a) A stream of fast-moving electrons.
   (b) A form of electromagnetic radiation.
   (c) Highly ionizing than γ-radiation.
   (d) More penetrating than β-radiation.
2. A radioactive nuclide emits a β-particle. The atomic number (proton number) of the nucleus
   (a) Stays the same. (b) Increases by 1.
   (c) Decreases by 2. (d) Decreases by 4.
3. A radioactive element emits a particle from the nucleus of one of its atoms. The particle comprises two protons and two neutrons. The name of this process is called
   (a) α-emission (b) β-emission
   (c) γ-emission (d) Nuclear fission
4. A radioactive decay can be represented as shown. ${}_{91}Pa^{233} \rightarrow {}_{92}U^{233} + \ldots\ldots$
   The emitted particle is a/an
   (a) Gamma-ray. (b) Proton.
   (c) α-particle. (d) β-particle.
5. The type of radiation that travels in a straight line across an electric field is a/an
   (a) Proton (b) Electron
   (c) Alpha particle (d) Gamma-ray
6. A powder contains 100mg of a radioactive material that emits α-particles. The half-life of the isotope is five days. The mass of isotope that remains after ten days will be
   (a) 0mg (b) 25mg (c) 50mg (d) 75mg
7. The main source of energy in the stars is.
   (a) Chemical reaction (b) Nuclear fission
   (c) Nuclear fusion (d) Mechanical energy
8. The splitting of a heavy nucleus into smaller nuclei is called
   (a) Fusion (b) Fission
   (c) Half-life (d) Gamma decay
9. A process in which two light nuclei combine to form a heavier nucleus is called
   (a) Nuclear fusion (b) Nuclear fission
   (c) Beta-decay (d) Alpha-decay



10. Which row shows the nature and the penetrating ability of β-particles?

	Nature	Most are stopped by
a)	helium nucleus	a few mm of aluminum
b)	helium nucleus	a thin sheet of paper
c)	electron	a few mm of aluminum
d)	electron	a thin sheet of paper

11. Compared with α -particles and β – particles, γ- rays,
(a) Are a type of radiation to carry a charge.
(b) Have the most significant ionizing effect.
(c) Have the most significant penetrating effect.
(d) Have the most negligible mass.

12. The severe health hazards caused by radioactive emissions is/are.
(a) Cancer (b) Genetic change
(c) Deep-sited burns (d) All of these

13. Radioactive materials should be handled carefully. Which safety measure does not reduce the risk of using radioactive material?
(a) Keeping the material a long distance
(b) Keeping the material at a low temperature
(c) Using lead screening
(d) Using the material for a short time

14. A scientist experiments using a sealed source that emits β-particles. The range of the β-particles in the air is about 30cm. The precaution that is the most effective to protect the scientist from the radiation is,
(a) Handling the source with long tongs
(b) Keeping the temperature of the source low
(c) Opening all windows in the laboratory
(d) Washing his hands before leaving the laboratory

15. The safest way to dispose of a large quantity of radioactive waste is,
(a) Burying it in a dry rock deep underground
(b) Washing it in the drain
(c) Burning it on a fire
(d) Draining it into the sea

Section (B) Structured Questions

1. a) Define the term radioactivity.
b) What do you mean by the term stable nucleus?
c) Why are some elements radioactive, but some are not?